Uniformly most split reliable graphs: a complete answer outside two edge counts*

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Abstract

A network with two distinguished terminals is split successfully when independent edge failures leave exactly two connected components, one containing each terminal. We ask when a connected simple graph with fixed numbers of vertices and edges maximises this probability for every common edge-retention probability. For $n \geq 7$, we prove nonexistence for $n \leq m \leq \binom{n}{2} - n - 1$ and existence for $m \geq \binom{n}{2} - n + 2$; trees admit an optimiser by earlier work. Only two edge counts per order remain conditional. We reduce those cases to an explicit family of cone-reliability inequalities and certify the required inequalities in exact integer arithmetic through cone order 50, completing the existence classification for $7 \leq n \leq 51$. The structural argument combines a prior classification of local optimisers with explicit spanning-tree competitors and compatible extrema in the identity $\mathrm{SR}_{s,t}(H) = \mathrm{Rel}(H/st) - \mathrm{Rel}(H)$. We also prove the boundary cone comparison at every order for $p \geq 1/2$ and on an explicit positive interval near zero. The intervening probability range remains open at unrestricted orders. We distinguish the universal written proofs, the finite certificates needed for the boundary result, and supplementary computational corroboration.

1 Introduction

1.1 Why this optimisation problem is different

Ordinary network reliability rewards keeping every vertex connected. Here the desired outcome is different: two distinguished sources must remain separated, but every other vertex must still be connected to exactly one of them. A surviving network with three components fails this requirement, as does one connected component containing both sources. The model is an idealised independent-edge failure model, not a validated design prescription for a physical network.

The word *uniformly* is the demanding part. We fix both the number of vertices and the number of edges, and ask whether one design is best for **every** common edge-retention probability. A design that wins when most edges survive can lose when most fail. Consequently, nonexistence of a uniform optimiser does not mean that no best design exists at a specified probability; it means that the best choice cannot be made independently of that probability.

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Our proof separates these competing endpoint requirements. Near complete edge failure, spanning two-forests determine the first nonzero term. Near perfect edge retention, a prior classification forces a particular split polynomial. In the intermediate density range these requirements conflict. In the dense range, identifying the two sources converts the comparison into two compatible all-terminal reliability comparisons.

1.2 Problem and prior results

Let H be a connected simple graph with n vertices and m edges and let $s \neq t$ be two of its vertices, the terminals. Each edge is deleted independently with probability $1-p$. The *all-terminal reliability* $\mathrm{Rel}(H;p)$ is the probability that the surviving spanning subgraph is connected. The *split reliability* $\mathrm{SR}_{s,t}(H;p)$, introduced by Brown and Mol (2017) and studied as a uniform optimisation problem by Brown and McMullin (2023), is the probability that the surviving spanning subgraph has exactly two connected components, one containing s and the other containing t: every vertex can still communicate with exactly one of the two terminals. Writing $N_i = N_i(H)$ for the number of such *split states* with exactly i surviving edges,

$$\mathrm{SR}_{s,t}(H;p) = \sum_{i=n-2}^{m-1} N_i(H)\, p^i (1-p)^{m-i}.$$

Let $\mathcal{T}_{n,m}$ be the class of two-terminal connected simple graphs with n vertices and m edges. A member G is *uniformly most split reliable* (UMSR) in $\mathcal{T}_{n,m}$ if $\mathrm{SR}_G(p) \geq \mathrm{SR}_H(p)$ for every $H \in \mathcal{T}_{n,m}$ and every $p \in [0,1]$. Brown and McMullin asked for which pairs (n,m) such a graph exists. They showed that a UMSR graph always exists for $m = n-1$ (the path with terminals at its ends), for $m = \binom{n}{2}$ and for $m = \binom{n}{2} - 1$, that none exists for $m = n \geq 6$, and by computation that for $n \leq 5$ one exists for every m, for $n = 6$ one exists unless $m \in \{6, 8\}$, and for $n = 7$ one exists if and only if $m = 6$ or $14 \leq m \leq 21$. The remaining cases were left open. Romero (2025) characterised the *locally* most split reliable graphs (those optimal for p close to 1): they are exactly the two-terminal graphs with the same split reliability polynomial as a two-terminal balloon graph $G_{n,m}$. Consequently any UMSR graph has the split polynomial of $G_{n,m}$. He proved that no UMSR graph exists when $n \geq 7$ and $n \leq m \leq \binom{n-3}{2} + 3$, and asked about the range $\binom{n-3}{2} + 4 \leq m \leq \binom{n}{2} - 2$.

Our main result answers this for all but two values of m for each n, and for those two values reduces the question to a well known conjecture in all-terminal reliability.

Theorem 1.1 (Main theorem)**.** *Let $n \geq 7$ and $n-1 \leq m \leq \binom{n}{2}$, and write $k = \binom{n}{2} - m$ for the number of missing edges.*

- *(i) If $m = n-1$ then a UMSR graph exists (the path with the terminals at its ends).*
- *(ii) If $n \leq m \leq \binom{n}{2} - n - 1$, equivalently $n+1 \leq k \leq \binom{n}{2} - n$, then no UMSR graph exists in $\mathcal{T}_{n,m}$.*
- *(iii) If $m \geq \binom{n}{2} - n + 2$, equivalently $k \leq n-2$, then the two-terminal balloon graph $G_{n,m}$ (Definition 2.1) is UMSR.*
- *(iv) If $m \in \{\binom{n}{2} - n, \binom{n}{2} - n + 1\}$, equivalently $k \in \{n-1, n\}$, then $G_{n,m}$ is UMSR provided the balloon graph $B_{n,m}$ is uniformly least reliable among connected graphs with n vertices and m edges (in the all-terminal sense). This proviso holds for $7 \leq n \leq 51$, and for every $n \geq 11$ it follows from inequality $(\star)_{n-1,\varepsilon}$ of the boundary-reduction subsection (Theorem 4.5).*

In particular, for $7 \leq n \leq 51$ a UMSR graph exists in $\mathcal{T}_{n,m}$ if and only if $m = n-1$ or $m \geq \binom{n}{2} - n$.

For $7 \le n \le 51$, the classification of vertex-and-edge counts is simple: a uniform optimiser exists precisely for trees or when at most n edges are missing. For larger orders the two boundary edge counts remain conditional. This is an existence classification, not a claim that every graph in an admissible class is optimal. Part (ii) is new for $\binom{n-3}{2} + 4 \le m \le \binom{n}{2} - n - 1$ (for smaller m it is Romero's Theorem 12). Part (iii) is new for $k \ge 2$. The two edge counts in (iv) are the two values immediately below the known dense range for the all-terminal "uniformly least reliable graph" question: Petingi, Saccoman and Schoppmann (1996) proved that balloon graphs are uniformly least reliable for $m \ge \binom{n-1}{2} + 1$ (that is, $k \le n - 2$), and Romero (2022) proved the conjecture of Boesch, Satyanarayana and Suffel for $m \le n + 3$; the values $m = \binom{n-1}{2}$ and $\binom{n-1}{2} - 1$ lie just below the dense range. Our reduction shows that progress on that conjecture at these two edge counts immediately completes the split reliability picture, and Theorem 4.5 reduces the conjecture at these two edge counts, for all n, to a family of inequalities between explicit cone graphs, which we verify exactly for $n \le 51$.

The reader can follow three proof stages. First, the preliminaries establish the identification identity and exchange tool. Next, the nonexistence argument uses Romero's candidate theorem and two explicit competitors. Finally, the dense-existence argument combines identified-graph maximality with an imported all-terminal minimum theorem, then isolates the two boundary cases. The computation section distinguishes the exact finite certificates from supplementary checks. Readers interested chiefly in the scope can read the main theorem, the small triangle example, and the final discussion before the technical proofs.

2 Preliminaries

The following notation separates three objects that must not be conflated: the original two-terminal graph, its terminal identification, and the ordinary balloon graph used in a least-reliability theorem.

Symbol	Meaning	Role in the proof
H, s, t	Connected simple graph and its two distinct terminals	Competing network
n, m, k	Vertices, edges, and $k = \binom{n}{2} - m$ missing edges	Fixed comparison class
p, q	Edge retention and failure probabilities, $q = 1 - p$	Uniform comparison variable
H/st	Terminal identification, with parallel edges retained	First term of the identity
$B_{n,m}$	Ordinary balloon graph	All-terminal minimisation
$G_{n,m}$	Balloon with diameter terminals	Split-reliability candidate
N_i, c_i	Split-state and connected-state edge counts	Exact finite certificates
$C_N(s)$	Clique on $N - 1$ vertices plus a vertex with s neighbours	Boundary reduction

2.1 Notation

Graphs are finite; simple unless stated otherwise. For a loopless multigraph Y we write $\mu_Y(uv)$ for the number of uv edges and $e_v(U) = \sum_{u \in U} \mu_Y(vu)$ for the number of edges (with multiplicity) from

v into a vertex set U. K_q is the complete graph, $K_{1,a}$ the star with a leaves, K_2 a single edge, and $K_q - F$ denotes K_q with the edge set of a graph F on (a subset of) its vertices removed. $\tau(Y)$ is the number of spanning trees of Y and $c_i(Y)$ the number of connected spanning subgraphs with exactly i edges, so that $\mathrm{Rel}(Y;p) = \sum_i c_i(Y)p^i(1-p)^{|E(Y)|-i}$ and $\mathrm{Rel}(Y;p) = \tau(Y)p^{|V(Y)|-1} + O(p^{|V(Y)|})$ as $p \to 0$. The *two-terminal* version of a graph carries an unordered pair $\{s,t\}$ of distinct terminals; the split polynomial is symmetric in s and t.

A *split-(s,t)-bridge* of H is an edge whose deletion leaves s and t in different components. Since $N_{m-1}(H)$ counts single edges whose deletion produces a split state, $N_{m-1}(H)$ equals the number of split-(s,t)-bridges.

For a two-terminal graph H we write H/st for the multigraph obtained by identifying s and t into a single vertex z (an st edge becomes a loop, which plays no role for connectivity and is discarded). Thus $\mu_{H/st}(zw) = [sw \in E(H)] + [tw \in E(H)] \in \{0,1,2\}$ for $w \neq s,t$, and H/st is simple away from z.

2.2 Balloon graphs and the candidate $G_{n,m}$

Definition 2.1 (Balloon graphs, after Romero 2025)**.** For $n \geq 4$ and $n \leq m \leq \binom{n}{2}$ the balloon graph $B_{n,m}$ is defined by: if $m \geq \binom{n-1}{2} + 2$, $B_{n,m}$ is K_{n-1} together with one further vertex joined to $m - \binom{n-1}{2}$ vertices of the clique; $B_{4,4}$ is a triangle with a pendant vertex; otherwise $B_{n,m}$ is $B_{n-1,m-1}$ with a new pendant vertex attached to a vertex of minimum degree. The two-terminal balloon graph $G_{n,m}$ is $B_{n,m}$ with terminals at distance equal to the diameter of $B_{n,m}$.

Romero (2025, Theorem 10) proves that $G_{n,m}$ is unique up to isomorphism and that the locally most split reliable graphs in $\mathcal{T}_{n,m}$ are exactly the two-terminal graphs with the same split polynomial as $G_{n,m}$ ("split-equivalent" to it). We shall only need the explicit form of $G_{n,m}$ in the ranges that matter, which is read off directly from the definition (with $k = \binom{n}{2} - m$):

- $0 \leq k \leq n-2$ (so $m \geq \binom{n-1}{2} + 1$): $G_{n,m} = K_n - K_{1,k}$ where the missing star is centred at s and t is one of its leaves (for $k = 0$ the terminals are any two vertices). Thus s has degree $n-1-k$, every other vertex is adjacent to everything except possibly s, and s,t are non-adjacent when $k \geq 1$.
- $k = n-1$: s is a pendant vertex attached to u, the other $n-1$ vertices induce K_{n-1} minus one edge ux, and $t = x$.
- $k = n$: s is a pendant vertex attached to u, the other $n-1$ vertices induce K_{n-1} minus the two edges ux, uy, and $t = x$.
- $n+1 \leq k \leq 2n-6$ (regime A1 below): s is pendant at u; the other $n-1$ vertices induce $K_{n-1} - K_{1,r}$ with the missing star centred at u, where $r = k - n + 2 \in [3, n-4]$; and t is a leaf of that star (a non-neighbour of u).
- $2n-5 \leq k \leq 3n-10$ (regime A2 below): s is the end of a pendant path $s-w-u$ of length two; the other $n-2$ vertices induce $K_{n-2} - K_{1,r''}$ with the missing star centred at u, where $r'' = k - 2n + 5 \in [0, n-5]$; and t is a leaf of that star (any vertex of the clique other than u when $r'' = 0$).

These descriptions specify the representatives used below. The proof uses equality of split polynomials, not uniqueness of all optimal two-terminal graphs.

2.3 Behaviour near $p = 0$ and $p = 1$

The following elementary observation is Lemma 3 of Romero (2025), attributed to Brown and McMullin.

Lemma 2.2. *Let $G, H \in \mathcal{T}_{n,m}$. If $N_i(G) = N_i(H)$ for all $i < i_0$ and $N_{i_0}(G) > N_{i_0}(H)$, then $\mathrm{SR}_G(p) > \mathrm{SR}_H(p)$ for all p in some interval $(0, \delta)$. If $N_{m-j}(G) = N_{m-j}(H)$ for all $j < j_0$ and $N_{m-j_0}(G) > N_{m-j_0}(H)$, then $\mathrm{SR}_G(p) > \mathrm{SR}_H(p)$ for all p in some interval $(1 - \delta, 1)$.*

Proof. $\mathrm{SR}_G - \mathrm{SR}_H = \sum_i \big(N_i(G) - N_i(H)\big) p^i (1-p)^{m-i}$, and the first non-vanishing term dominates near the relevant endpoint. □

Consequently a UMSR graph must maximise N_{n-2} over $\mathcal{T}_{n,m}$ and must maximise the vector $(N_{m-1}, N_{m-2}, \dots)$ in lexicographic order. Nonexistence is proved by showing that these two requirements are incompatible.

2.4 Split reliability as a difference of reliabilities

Lemma 2.3. *For every two-terminal graph H and every p,*

$$\mathrm{SR}_{s,t}(H; p) = \mathrm{Rel}(H/st; p) - \mathrm{Rel}(H; p),$$

where $\mathrm{Rel}(H/st; p)$ is the probability that the surviving edge set of H is connected after s and t are identified. In particular $N_{n-2}(H) = \tau(H/st)$.

Proof. A spanning subgraph of H is connected in H/st if and only if it is connected in H, or it has exactly two components with s and t in different ones. The two events are disjoint, and the second is the split event. The last claim compares the coefficients of p^{n-2} as $p \to 0$: H/st has $n-1$ vertices, so $\mathrm{Rel}(H/st; p) = \tau(H/st) p^{n-2} + O(p^{n-1})$, whereas $\mathrm{Rel}(H; p) = O(p^{n-1})$. □

The second statement is the classical fact that the number of spanning 2-forests of H separating s from t equals the number of spanning trees of H with s and t identified. We shall use it in both directions: in Section 3 to evaluate N_{n-2} of the candidates and competitors through spanning-tree counts of dense graphs, and in Section 4 to transfer the whole polynomial comparison to all-terminal reliability.

A three-vertex example and the loop convention. Take a triangle with terminals s, t and third vertex v. A split state consists of exactly one of sv, tv: the edge st must be absent. Thus its split reliability is $2p(1-p)^2$. Identification leaves two parallel edges from the merged vertex to v, with reliability $1 - (1-p)^2 = 2p - p^2$; the triangle itself has reliability $3p^2 - 2p^3$. Subtracting gives $2p - 4p^2 + 2p^3$, as required.

Discarding the identified loop does not change a probability, but it does change edge-count coefficients. If X denotes the loopless identification, with $c_i = 0$ outside its coefficient range, the exact identity is

$$N_i(H) = c_i(X) + \mathbf{1}_{st \in E(H)} c_{i-1}(X) - c_i(H).$$

The extra term counts states in which the discarded loop is selected. In generating-polynomial form its contribution is multiplication by $1 + x$. All coefficient comparisons below use a common total edge count; silently subtracting coefficient vectors of different edge counts is not valid.

2.5 Spanning trees of nearly complete graphs

Lemma 2.4. *Let F be a graph on a subset of the vertices of K_q, with Laplacian L_F (as a matrix indexed by the vertices of F). Then*

$$\tau(K_q - F) = q^{\,q-2} \det(I - \tfrac{1}{q} L_F).$$

Consequently, for $a \geq 1$,

$$\tau(K_q - K_{1,a}) = (q-a-1)(q-1)^{a-1} q^{\,q-a-2}, \qquad \tau\big(K_q - (K_{1,a} \cup K_2)\big) = (q-a-1)(q-1)^{a-1}(q-2)\, q^{\,q-a-3},$$

where in the second formula the star and the edge are vertex-disjoint.

Proof. $L(K_q - F) = qI - J - L_F$, and J (the all-ones matrix) commutes with L_F because L_F has zero row sums. Hence the two matrices are simultaneously diagonalisable, the nonzero eigenvalues of $L(K_q - F)$ are $q - \mu$ for μ running over the Laplacian eigenvalues of F (extended by zeros to all q vertices) with one zero removed, and by the matrix-tree theorem $\tau(K_q - F) = \frac{1}{q} \prod (q - \mu) = \frac{1}{q^2} \det(qI - L_F) = q^{q-2} \det(I - \frac{1}{q} L_F)$. The Laplacian eigenvalues of $K_{1,a}$ are $a+1$, 1 (with multiplicity $a-1$) and 0; those of K_2 are 2 and 0; the displayed formulas follow. □

2.6 A multigraph exchange lemma

For a loopless multigraph Y and distinct vertices x, y, z let $P_Y(xy \,|\, z)$ denote the probability (edges surviving independently with probability p) that the surviving spanning subgraph has exactly two connected components, one containing x and y and the other containing z; let $P_Y(x \,|\, y)$ denote the probability of exactly two components with x and y in different ones. For an edge xy (possibly already present) write $Y + xy$ for Y with one additional xy edge. Adding the edge xy makes a surviving edge set connected exactly when it was connected or had exactly two components separating x from y, so

$$\mathrm{Rel}(Y + xy; p) = \mathrm{Rel}(Y; p) + p\, P_Y(x \,|\, y). \qquad (2.1)$$

Lemma 2.5 (Exchange lemma)**.** *Let Y be a loopless multigraph and a, b, c distinct vertices. Suppose that*

$$e_b(U) \leq e_c(U) \qquad \text{for every } U \text{ with } a \in U \subseteq V(Y) \setminus \{b, c\}.$$

Then $P_Y(ab \,|\, c) \leq P_Y(ac \,|\, b)$ for every $p \in [0,1]$; consequently $P_Y(a \,|\, c) \leq P_Y(a \,|\, b)$ and $\mathrm{Rel}(Y + ab; p) \geq \mathrm{Rel}(Y + ac; p)$ for every p.

Proof. Let E_0 be the set of edges of Y incident with neither b nor c, and condition on the set T of surviving edges of E_0. The components of T partition $V' = V(Y) \setminus \{b, c\}$; call their vertex sets $U_1, \dots, U_r$ with $a \in U_1$. Given T, the events "b has a surviving edge into U_j" (probability $\alpha_j = 1 - (1-p)^{e_b(U_j)}$), "$c$ has a surviving edge into U_j" (probability $\beta_j = 1 - (1-p)^{e_c(U_j)}$), for $j = 1, \dots, r$, and "no bc edge survives" (probability $(1-p)^{\mu(bc)}$) concern pairwise disjoint sets of edges and are therefore independent. The event $\{ab \,|\, c\}$ occurs exactly when no bc edge survives, each U_j is joined to exactly one of b, c, and U_1 is joined to b; hence

$$P(ab \,|\, c \mid T) = (1-p)^{\mu(bc)}\, \alpha_1(1-\beta_1) \prod_{j \geq 2} \big(\alpha_j(1-\beta_j) + \beta_j(1-\alpha_j)\big),$$

and $P(ac \,|\, b \mid T)$ is the same expression with $\alpha_1(1-\beta_1)$ replaced by $\beta_1(1-\alpha_1)$. Since $\alpha_1(1-\beta_1) \leq \beta_1(1-\alpha_1)$ is equivalent to $\alpha_1 \leq \beta_1$, that is to $e_b(U_1) \leq e_c(U_1)$, which holds by hypothesis, the first claim follows by averaging over T. The second follows from $P_Y(a \,|\, c) = P_Y(ab \,|\, c) + P_Y(a \,|\, bc)$ and $P_Y(a \,|\, b) = P_Y(ac \,|\, b) + P_Y(a \,|\, bc)$, and the third from (2.1). □

This conditioning formulation belongs to the Kelmans/swing-surgery tradition (Kelmans, 1981; Satyanarayana, Schoppmann and Suffel, 1992). Brown, Colbourn and Devitt (1993) already extended reliability transformations to multigraphs, as explicitly recorded by Csikvári (2011, Section 2). We give a self-contained proof of the exact setwise condition used here, without claiming priority for the multigraph extension or a strict improvement over the 1993 hypotheses; the full 1993 text was not available for an exact comparison. The same conditioning identifies the leading coefficient as $p \to 0$:

Corollary 2.6. *Under the hypothesis of Lemma 2.5,*

$$\tau(Y + ab) - \tau(Y + ac) = \sum_{T} \big(e_c(U_1) - e_b(U_1)\big) \prod_{j \geq 2} \big(e_b(U_j) + e_c(U_j)\big) \ \geq 0,$$

the sum over the spanning forests T of $Y - \{b, c\}$, with $U_1, \ldots, U_r$ the vertex sets of the trees of T and $a \in U_1$. In particular, if $Y - \{b, c\}$ is connected and $\deg_Y(c) > \deg_Y(b)$ then $\tau(Y + ab) > \tau(Y + ac)$.

Proof. By (2.1), $\mathrm{Rel}(Y + ab) - \mathrm{Rel}(Y + ac) = p\big(P_Y(ac \,|\, b) - P_Y(ab \,|\, c)\big)$, and the difference of the two conditional probabilities in the proof of Lemma 2.5 is $(1-p)^{\mu(bc)}(\beta_1 - \alpha_1) \prod_{j \geq 2}(\cdots)$. As $p \to 0$, a given edge set T with r components has probability $p^{|T|}(1 + O(p))$, which is $p^{|V'|-r}(1 + O(p))$ when T is a forest and is of higher order otherwise; $\beta_1 - \alpha_1 = p\,(e_c(U_1) - e_b(U_1)) + O(p^2)$; and each factor $\alpha_j(1-\beta_j) + \beta_j(1-\alpha_j) = p\,(e_b(U_j) + e_c(U_j)) + O(p^2)$. Collecting terms, $\mathrm{Rel}(Y + ab) - \mathrm{Rel}(Y + ac) = p^{|V'|+1} \sum_T (\cdots) + O(p^{|V'|+2})$, and $|V'| + 1 = |V(Y)| - 1$ is the exponent of p carrying τ in Rel. Each term of the sum is nonnegative by hypothesis; a spanning tree T of $Y - \{b, c\}$ contributes $e_c(V') - e_b(V') = \deg_Y(c) - \deg_Y(b)$. □

3 Nonexistence in the remaining intermediate-density range

We use Romero's (2025, Theorem 10) characterization of locally most split reliable graphs: for the parameters considered here, every such graph is split-equivalent to the canonical balloon candidate $G_{n,m}$. In particular, a UMSR graph H must satisfy

$$\mathrm{SR}_H = \mathrm{SR}_{G_{n,m}}, \qquad N_{n-2}(H) = N_{n-2}(G_{n,m}).$$

Consequently it suffices to construct a connected simple two-terminal graph with the same numbers of vertices and edges and a strictly larger N_{n-2} coefficient. Such a graph beats $G_{n,m}$ for all sufficiently small positive p. This argument concerns split-equivalence, not uniqueness up to graph isomorphism.

3.1 Regime A1: one pendant terminal

Proposition 3.1. *Let $n \geq 7$, $n + 1 \leq k \leq 2n - 6$, $m = \binom{n}{2} - k$, and $r = k - n + 2$. No graph in $\mathcal{T}_{n,m}$ is UMSR.*

Proof. Here $3 \leq r \leq n - 4$. The canonical candidate consists of a core $C = K_{n-1} - K_{1,r}$, with missing star centred at u, and a pendant terminal s at u; the other terminal x is a missing-star leaf. Identifying s and x restores ux, so

$$N_{n-2}(G_{n,m}) = \tau(K_{n-1} - K_{1,r-1}).$$

Construct H' with a pendant terminal s at u, other terminal x, and core

$$H' - s = K_{n-1} - (K_{1,r-1} \cup K_2),$$

where the missing star is centred at u, contains x as a leaf, and is vertex-disjoint from the additional missing edge bc. The construction uses $r + 2 \leq n - 2$ core vertices. The core induced by vertices other than u is a clique minus bc and is connected; u has $n - 1 - r \geq 3$ neighbours in it. Thus H' is connected and simple. Its number of missing edges is $(n - 2) + (r - 1) + 1 = k$.

Identification restores ux. Applying the spanning-tree formulas of Lemma 2.4,

$$\frac{N_{n-2}(H')}{N_{n-2}(G_{n,m})} = \frac{\tau(K_{n-1} - (K_{1,r-2} \cup K_2))}{\tau(K_{n-1} - K_{1,r-1})} = \frac{(n-r)(n-3)}{(n-1-r)(n-2)} > 1,$$

since the numerator minus denominator of the last fraction is $r - 2 > 0$. The candidate reduction above proves the assertion. □

3.2 Regime A2: a pendant path of length two

The preceding exchange lemma supplies a strict spanning-tree comparison without requiring a closed formula for every missing-star size.

Proposition 3.2. *Let $n \geq 7$, $2n - 5 \leq k \leq 3n - 10$, $m = \binom{n}{2} - k$, and $r = k - 2n + 5$. No graph in $\mathcal{T}_{n,m}$ is UMSR.*

Proof. Put $q = n - 2$, so $0 \leq r \leq q - 3$. Define a core C on q vertices as follows. For $r \geq 1$, let $C = K_q - K_{1,r}$ with missing star centred at u and choose a missing leaf x. For $r = 0$, let $C = K_q$ and choose distinct vertices u, x. In particular, the zero-star case does not require a nonexistent missing leaf.

The canonical candidate is obtained by adjoining the path $s - w - u$ to C, with terminals s, x. Its identified graph $\Gamma = G_{n,m}/sx$ consists of C and the vertex w joined exactly to u, x.

Choose $\ell \in V(C) \setminus \{u, x\}$ adjacent to u in C. There are $q - r - 1 \geq 2$ choices when $r \geq 1$, and $q - 2 \geq 3$ when $r = 0$. Choose any $a \in V(C) \setminus \{u, x, \ell\}$. The edge $a\ell$ belongs to C, whereas aw is absent from Γ. Set

$$\Gamma' = \Gamma - a\ell + aw.$$

Form H' by deleting wx from Γ' and adjoining a new pendant vertex s at w, with terminals s, x. This is a simple graph with n vertices and $\binom{q}{2} - r + 2 = m$ edges. It is connected: the subgraph of $C - a\ell$ induced by $V(C) \setminus \{u\}$ is a clique minus one edge on $q - 1 \geq 4$ vertices, and u retains its $q - 1 - r \geq 2$ neighbours. Vertex w is joined to u and a, and s to w. Identification of s, x restores wx, giving $H'/sx = \Gamma'$.

To compare spanning-tree counts, put $Y = \Gamma - a\ell$. In Y, w has precisely the neighbours u, x, while ℓ is adjacent to every vertex other than w, a, ℓ. Therefore, for every $a \in U \subseteq V(Y) \setminus \{w, \ell\}$,

$$e_w(U) = |U \cap \{u, x\}| \leq |U| - 1 = e_\ell(U).$$

Moreover, $Y - \{w, \ell\} = C - \ell$ is connected: its vertices other than u form a clique, and deleting ℓ leaves at least $q - 2 - r \geq 1$ neighbours of u. Finally,

$$\deg_Y(\ell) = q - 2 = n - 4 > 2 = \deg_Y(w).$$

The strict spanning-tree corollary 2.6, with $(b, c) = (w, \ell)$, now gives

$$N_{n-2}(H') = \tau(Y + aw) > \tau(Y + a\ell) = \tau(\Gamma) = N_{n-2}(G_{n,m}).$$

This covers $r = 0$ as well as $r \geq 1$, and the candidate reduction proves the proposition. □

Theorem 3.3. *For $n \geq 7$ and $\binom{n-3}{2} + 4 \leq m \leq \binom{n}{2} - n - 1$, no graph in $\mathcal{T}_{n,m}$ is UMSR. Together with Romero (2025, Theorem 12), this gives nonexistence for every $n \leq m \leq \binom{n}{2} - n - 1$.*

Proof. The two propositions cover all missing-edge counts $n+1 \leq k \leq 3n-10$, precisely the displayed range of m. Romero's Theorem 12 covers $n \leq m \leq \binom{n-3}{2} + 3$. □

4 Existence for $m \geq \binom{n}{2} - n + 2$, and the two boundary values

4.1 The identified graph of a two-terminal graph

Fix $n \geq 4$, $1 \leq k \leq n$ and $m = \binom{n}{2} - k$. Let $d = n - 1 - k \geq -1$. Define the loopless multigraph $X^{(k)}$ on $n-1$ vertices as follows: for $k \leq n-1$, $X^{(k)}$ is K_{n-1} with d additional edges parallel to d distinct edges at one fixed vertex z (so $X^{(n-1)} = K_{n-1}$); and $X^{(n)} = K_{n-1} - e$. From the explicit description of $G_{n,m}$ in the balloon-graph definition one checks directly that $G_{n,m}/st \cong X^{(k)}$: for $k \leq n-2$ the merged vertex is joined twice to each of the $n-1-k$ neighbours of s (all of which are neighbours of t) and once to every other vertex; for $k = n-1$ the neighbourhoods of s and t are disjoint and together cover the other vertices; for $k = n$ the same holds except that the edge uy is missing from the clique.

Lemma 4.1. *For every $H \in \mathcal{T}_{n,m}$ and every $p \in [0,1]$, $\mathrm{Rel}(H/st; p) \leq \mathrm{Rel}(X^{(k)}; p)$.*

Proof. Let $W = V(H) \setminus \{s, t\}$ and let $\mathfrak{X}$ be the family of loopless multigraphs X on $W \cup \{z\}$ with exactly m edges such that $X[W]$ is simple and $\mu(zw) \leq 2$ for all $w \in W$. If $st \notin E(H)$ then $H/st \in \mathfrak{X}$. If $st \in E(H)$, then H/st (with its loop discarded) has $m-1$ edges, and since $k \geq 1$ some edge of K_n other than st is missing from H; adding the corresponding edge (a W-edge, or one more copy of a z-edge) produces a member X of $\mathfrak{X}$ with $\mathrm{Rel}(H/st) \leq \mathrm{Rel}(X)$ by monotonicity. So it suffices to prove $\mathrm{Rel}(X) \leq \mathrm{Rel}(X^{(k)})$ for $X \in \mathfrak{X}$.

Describe $X \in \mathfrak{X}$ by the set F_W of pairs of W not joined in X ($f_W = |F_W|$), the set $Z_0 = \{w : \mu(zw) = 0\}$ and the set $D_2 = \{w : \mu(zw) = 2\}$. Counting edges,

$$|D_2| = d + f_W + |Z_0|. \qquad (4.2)$$

Call X *terminal* if $f_W = 0$ and $Z_0 = \emptyset$ when $d \geq 0$, and if $f_W + |Z_0| = 1$ and $D_2 = \emptyset$ when $d = -1$. A terminal X is isomorphic to $X^{(k)}$: for $d \geq 0$ it is K_{n-1} with exactly d doubled edges at z by (4.2); for $d = -1$ it is K_{n-1} minus one edge. We show that every non-terminal $X \in \mathfrak{X}$ admits a move to some $X' \in \mathfrak{X}$ with $\mathrm{Rel}(X') \geq \mathrm{Rel}(X)$ that strictly decreases $\Phi(X) = 2f_W + |Z_0|$; since $\Phi \geq 0$ the process ends at a terminal graph, proving the lemma.

Move M1 (fill a missing W-edge). Let $xy \in F_W$ with $\mu(zx) \geq 1$ and $|Z_0 \cap N_W(y)| \leq \mu(zx) - 1$, where $N_W(y)$ is the set of W-neighbours of y in X. Let Y be X with one copy of zx removed and $X' = Y + xy$. For U with $x \in U \subseteq W \setminus \{y\}$,

$$e_z(U) - e_y(U) = (\mu_X(zx) - 1) + \sum_{u \in U \setminus \{x\}} (\mu_X(zu) - [uy \in E(X)]) \;\geq\; \mu_X(zx) - 1 - |Z_0 \cap N_W(y)| \;\geq 0,$$

so Lemma 2.5 with $(a,b,c)=(x,y,z)$ gives $\mathrm{Rel}(X')=\mathrm{Rel}(Y+xy)\geq\mathrm{Rel}(Y+xz)=\mathrm{Rel}(X)$. The move removes xy from F_W and adds x to Z_0 only if $\mu_X(zx)=1$, so Φ decreases by at least 1.

Move M3 (fill a missing z-edge). Let $w\in Z_0$ and $v\in D_2$ with at most one F_W-neighbour of v outside $N_{F_W}(w)\cup\{w\}$, where N_{F_W} denotes neighbourhoods in the graph F_W. Let Y be X with one copy of zv removed and $X'=Y+zw$. For U with $z\in U\subseteq V(X)\setminus\{v,w\}$, the contribution of z to $e_v(U)-e_w(U)$ is $\mu_Y(vz)-\mu_Y(wz)=1$, and each $u\in U\setminus\{z\}$ contributes $[uv\in E]-[uw\in E]=[uw\in F_W]-[uv\in F_W]\geq -1$, with -1 exactly when $uv\in F_W$ and $uw\notin F_W$. Hence $e_v(U)-e_w(U)\geq 1-1=0$, and Lemma 2.5 with $(a,b,c)=(z,w,v)$ gives $\mathrm{Rel}(X')\geq\mathrm{Rel}(X)$. The move removes w from Z_0 and v from D_2, so Φ decreases by 1.

A move always exists. Suppose X is non-terminal. If $Z_0=\emptyset$ then $f_W\geq 1$ (for $d\geq 0$) or $f_W\geq 2$ (for $d=-1$), and M1 applies to any $xy\in F_W$ because its hypothesis reduces to $\mu(zx)\geq 1$. If $Z_0\neq\emptyset$, then $D_2\neq\emptyset$ by (4.2) (for $d=-1$ non-terminality gives $f_W+|Z_0|\geq 2$). If some $v\in D_2$ has F_W-degree at most 1, M3 applies with any $w\in Z_0$. Otherwise every $v\in D_2$ has F_W-degree at least 2, so $2f_W\geq\sum_{v\in D_2}\deg_{F_W}(v)\geq 2|D_2|=2(d+f_W+|Z_0|)$, forcing $d=-1$, $|Z_0|=1$, and equality throughout: every edge of F_W joins two vertices of D_2 of F_W-degree exactly 2. Then $f_W\geq 1$ by non-terminality, and for any $xy\in F_W$ we have $\mu(zx)=2$ and $|Z_0\cap N_W(y)|\leq 1$, so M1 applies. □

4.2 Proof of parts (iii) and (iv)

Theorem 4.2. *Let $n\geq 4$, $0\leq k\leq n-2$ and $m=\binom{n}{2}-k$. Then $G_{n,m}$ is uniformly most split reliable in $\mathcal{T}_{n,m}$.*

Proof. For $k=0$ there is nothing to prove. For $k\geq 1$, by Lemma 2.3, Lemma 4.1, and the isomorphism $G_{n,m}/st\cong X^{(k)}$,

$$\mathrm{SR}_H(p)=\mathrm{Rel}(H/st;p)-\mathrm{Rel}(H;p)\leq\mathrm{Rel}(G_{n,m}/st;p)-\mathrm{Rel}(H;p).$$

As an unlabelled graph $G_{n,m}$ is the balloon graph K_{n-1} plus a vertex of degree $n-1-k\geq 1$, and $m\geq\binom{n-1}{2}+1$; by the theorem of Petingi, Saccoman and Schoppmann (1996) this graph is uniformly least reliable among connected graphs with n vertices and m edges, so $\mathrm{Rel}(H;p)\geq\mathrm{Rel}(G_{n,m};p)$ for all p. Hence $\mathrm{SR}_H(p)\leq\mathrm{SR}_{G_{n,m}}(p)$ for all p, which is what the definition requires. □

Theorem 4.3. *Let $n\geq 7$ and $k\in\{n-1,n\}$, $m=\binom{n}{2}-k$. If $B_{n,m}$ is uniformly least reliable among connected graphs with n vertices and m edges, then $G_{n,m}$ is uniformly most split reliable in $\mathcal{T}_{n,m}$. The hypothesis holds for $7\leq n\leq 51$ (Corollary 4.7).*

Proof. The displayed inequality in the proof of Theorem 4.2 holds verbatim with Lemma 4.1 for $k\in\{n-1,n\}$, and $G_{n,m}$ is $B_{n,m}$ as an unlabelled graph; the hypothesis supplies $\mathrm{Rel}(H)\geq\mathrm{Rel}(G_{n,m})$. □

The hypothesis of Theorem 4.3 is only needed for graphs of minimum degree at least 2: if H has a pendant vertex then $\mathrm{Rel}(H)=p\,\mathrm{Rel}(H')$ where H' has $n-1$ vertices and $m-1\geq\binom{n-2}{2}+1$ edges, and the Petingi–Saccoman–Schoppmann theorem applied to H' gives $\mathrm{Rel}(H)\geq\mathrm{Rel}(B_{n,m})$. The rest of this section reduces the hypothesis, for all n, to an explicit family of inequalities between reliabilities of cones over a complete graph, and verifies that inequality exactly for $n\leq 51$.

4.3 Reduction of the hypothesis to an inequality between two cones

For $N \geq 3$ and $0 \leq s \leq N-1$ let $C_N(s) = K_N - K_{1,N-1-s}$ denote the *cone graph*: a vertex u joined to s vertices of a clique K_{N-1}. Thus $K_N - K_{1,1+\varepsilon} = C_N(N-2-\varepsilon)$, and $B_{n,m}$ for $m = \binom{n-1}{2} - \varepsilon$ is $C_{n-1}(n-3-\varepsilon)$ with a pendant vertex attached to a clique vertex. Write $f_N(s;p) = \mathrm{Rel}(C_N(s);p)$ and $q = 1-p$.

Definition 4.4 (Inequality $(\star)_{N,\varepsilon}$)**.** Let $N \geq 7$ and $\varepsilon \in \{0,1\}$, and put $h = N-2-\varepsilon$. We say that $(\star)_{N,\varepsilon}$ holds if for every integer d with $1 \leq d \leq N-5-\varepsilon$ and every $p \in [0,1]$,

$$(1-q^{d+1})\, f_N(h-d;p) \;\geq\; p\, f_N(h;p). \qquad (4.3)$$

Both sides of (4.3) are reliabilities of multigraphs on $N+1$ vertices with $\binom{N-1}{2} + h + 1$ edges: the right side is the balloon ($C_N(h)$ with a pendant vertex w attached to a clique vertex x), the left side is $C_N(h-d)$ with w attached to x by $d+1$ parallel edges. So $(\star)$ says that taking d of the cone edges away from u and giving them to the pendant vertex, as parallel copies of its one edge, does not decrease reliability. Expanding $1-q^{d+1} = \sum_{i\geq 1}\binom{d+1}{i}p^i q^{d+1-i}$, (4.3) is implied by the coefficient-wise statement

$$\sum_{i=1}^{d+1} \binom{d+1}{i}\, c_{j-i}(C_N(h-d)) \;\geq\; c_{j-1}(C_N(h)) \qquad \text{for all } j, \qquad (4.4)$$

because the basis $p^j q^{M-j}$ is nonnegative on $[0,1]$. Here $c_j(G)$ is the number of connected spanning subgraphs of G with j edges.

The reduction uses Kelmans's transformation. For two vertices u, v of a graph G, the *Kelmans transformation* with beneficiary u and co-beneficiary v deletes all edges vw with $w \in N(v)\backslash(N(u)\cup\{u\})$ and adds the edges uw for the same w; it preserves the number of edges. Kelmans (1981) proved that it never increases all-terminal reliability (see Csikvári, 2011, Theorem 2.2, with $k=1$ there), and Csikvári (2011, Theorem 2.6) proved that every graph can be carried by a sequence of Kelmans transformations to a threshold graph with the same number of components. Recall that a graph is *threshold* if any two vertices u, v have comparable sets $N(u)\setminus\{v\}$ and $N(v)\setminus\{u\}$; equivalently it is obtained from a single vertex by repeatedly adding an isolated or a dominating vertex. The complement of a threshold graph is threshold, and a threshold graph without isolated vertices has a vertex adjacent to all others.

Theorem 4.5. *Let $\varepsilon \in \{0,1\}$, $n \geq 8+3\varepsilon$, $m = \binom{n-1}{2} - \varepsilon$ and $N = n-1$. If $(\star)_{N,\varepsilon}$ holds, then every connected graph H with n vertices and m edges satisfies*

$$\mathrm{Rel}(H;p) \;\geq\; \mathrm{Rel}(B_{n,m};p) = p\,\mathrm{Rel}(K_{n-1} - K_{1,1+\varepsilon};p) \qquad (0 \leq p \leq 1),$$

i.e. $B_{n,m}$ is uniformly least reliable, and consequently $G_{n,m}$ is uniformly most split reliable in $\mathcal{T}_{n,m}$.

Proof. By the results of Kelmans and Csikvári just quoted there is a connected threshold graph H_{tr} with n vertices and m edges and $\mathrm{Rel}(H_{\mathrm{tr}}) \leq \mathrm{Rel}(H)$ on $[0,1]$, so we may assume that H itself is threshold. Let F be the complement of H; it is a threshold graph with $n-1+\varepsilon$ edges. Since H is connected and threshold, its last vertex in a threshold construction sequence is dominating, so F has an isolated vertex. Let t be the number of non-isolated vertices of F and F_0 the subgraph they induce; F_0 is threshold without isolated vertices, so it has a vertex w adjacent in F to the other $t-1$ vertices of F_0. In H the neighbours of w are exactly the $n-t$ vertices outside F_0, and each of them is isolated in F, hence dominating in H. Because $\binom{t}{2} \geq e(F_0) = n-1+\varepsilon$ and $n \geq 8+3\varepsilon$, we have $t \geq 5+\varepsilon$ (for $\varepsilon=0$, $t \leq 4$ would give $n-1 \leq 6$; for $\varepsilon=1$, $t \leq 5$ would give $n \leq 10$). Also $t \leq n-1$.

If $t = n-1$ then w is pendant, $\mathrm{Rel}(H) = p\,\mathrm{Rel}(H-w)$, and $H-w$ is a connected graph with $n-1$ vertices and $\binom{n-1}{2} - 1 - \varepsilon \geq \binom{n-2}{2} + 1$ edges, so by the Petingi–Saccoman–Schoppmann theorem $\mathrm{Rel}(H-w) \geq \mathrm{Rel}(K_{n-1} - K_{1,1+\varepsilon})$ and we are done.

Otherwise put $d = n-t-1$, so that w has degree $d+1 \geq 2$ and $1 \leq d \leq n-6-\varepsilon = N-5-\varepsilon$. The graph H is connected whenever $H-w$ is connected and at least one of the $d+1$ edges at w is present, and these two events concern disjoint edge sets; hence

$$\mathrm{Rel}(H) \;\geq\; (1-q^{d+1})\,\mathrm{Rel}(H-w).$$

The graph $H-w$ is connected (any neighbour of w dominates it), has N vertices, and has $(n-1+\varepsilon) - (t-1) = d+1+\varepsilon$ missing edges, i.e. $\binom{N}{2} - (d+1+\varepsilon) \geq \binom{N-1}{2} + 1$ edges because $d \leq N-3-\varepsilon$. By the Petingi–Saccoman–Schoppmann theorem,

$$\mathrm{Rel}(H-w) \;\geq\; \mathrm{Rel}\big(K_N - K_{1,d+1+\varepsilon}\big) = f_N(N-2-\varepsilon-d) = f_N(h-d).$$

Combining the two displays with $(\star)_{N,\varepsilon}$ gives $\mathrm{Rel}(H) \geq (1-q^{d+1}) f_N(h-d) \geq p\, f_N(h) = p\,\mathrm{Rel}(K_{n-1} - K_{1,1+\varepsilon})$. The last assertion is Theorem 4.3. □

The next proposition separates finite certification from an analytic endpoint comparison. The first excluded value has five small-order exceptions; it does not fail uniformly over all orders.

Proposition 4.6. *Let $\varepsilon \in \{0,1\}$, $N \geq 7$ and $h = N-2-\varepsilon$.*

- *(a) For $7 \leq N \leq 50$, the coefficient-wise inequality (4.4) holds for every $1 \leq d \leq N-5-\varepsilon$ and every j; in particular $(\star)_{N,\varepsilon}$ holds for $7 \leq N \leq 50$.*
- *(b) For every $N \geq 7$ and $1 \leq d \leq N-5-\varepsilon$, (4.3) holds to leading order as $p \to 0$: $(d+1)\,\tau(C_N(h-d)) \geq \tau(C_N(h))$.*
- *(c) At the first excluded value $d = N-4-\varepsilon = h-2$, (4.4) holds in the five cases $(N,\varepsilon) = (7,0),(7,1),(8,1),(9,1),(10,1)$. It fails in the other cases with $7 \leq N \leq 50$. For every $N \geq 11$, the analytic obstruction is already in the leading coefficient: $(d+1)\tau(C_N(h-d)) < \tau(C_N(h))$. Thus the proposed range cannot be enlarged uniformly in order.*

Proof. (a) is a finite computation in exact integer arithmetic, described in Section 5.

(b) By Lemma 2.4, $\tau(C_N(s)) = \tau(K_N - K_{1,N-1-s}) = s\,(N-1)^{N-2-s} N^{s-1}$, so

$$\frac{(d+1)\,\tau(C_N(h-d))}{\tau(C_N(h))} = \frac{(d+1)(h-d)}{h}\Big(1-\frac{1}{N}\Big)^{d}.$$

Put $s = h-d \geq 3$. The first factor $(d+1)s/(s+d)$ is increasing in s, so it is at least $3(d+1)/(d+3)$, and $(1-1/N)^d \geq e^{-d/(N-1)} \geq e^{-d/(d+4)}$ because $N-1 \geq h+1 \geq d+4$. The function $\varphi(d) = \log\frac{3(d+1)}{d+3} - \frac{d}{d+4}$ satisfies $\varphi(1) = \log 1.5 - 0.2 > 0$, $\varphi(d) \to \log 3 - 1 > 0$ as $d \to \infty$, and $\varphi'(d) = \frac{2}{(d+1)(d+3)} - \frac{4}{(d+4)^2}$ changes sign only once (from positive to negative), so $\varphi > 0$ for all $d \geq 1$; hence the ratio is at least 1.

(c) For $d = h-2$ the ratio in (b) equals $r(N,\varepsilon) = \frac{2(h-1)}{h}(1-\frac{1}{N})^{h-2}$. Since $h \geq N-3$, $r < 2(1-\frac{1}{N})^{N-5} \leq 2e^{-(N-5)/N}$, which is below 1 for $N \geq 17$; for $11 \leq N \leq 16$ direct evaluation gives $r \leq 0.988$ (the values at $N=11$ are 0.912 for $\varepsilon=0$ and 0.988 for $\varepsilon=1$). The excluded value is tested separately from (a). Exact coefficient checks give the five stated exceptions and failures in the remaining cases; these checks are retained separately in the evidence package. A negative leading coefficient proves failure near $p=0$, whereas an arbitrary negative coefficient alone would not suffice to refute the probability inequality. □

Corollary 4.7. *For $7 \le n \le 51$ and $m \in \{\binom{n-1}{2}, \binom{n-1}{2} - 1\}$ the balloon graph $B_{n,m}$ is uniformly least reliable among connected graphs with n vertices and m edges, and $G_{n,m}$ is uniformly most split reliable in $\mathcal{T}_{n,m}$. For every $n \ge 11$ the same conclusions follow from $(\star)_{n-1,\varepsilon}$ alone.*

Proof. The required base cases $7 \le n \le 10$ are covered by exact all-terminal coefficient comparisons in Section 5. For $8 \le n \le 51$ with $\varepsilon = 0$ and $11 \le n \le 51$ with $\varepsilon = 1$ it is Theorem 4.5 together with Proposition 4.6(a), applied with $N = n - 1 \le 50$. □

Theorem 4.5 replaces a statement about all connected graphs with n vertices and m edges by a statement about one explicit pair of graphs. The two sides of (4.3) have equal endpoint values, not equal first-order expansions. As $q \to 0$, their difference is $q - q^{d+1} - q^{h-d} + q^h + O(q^{N-2}) > 0$, and as $p \to 0$ it is governed by the spanning-tree ratio of Proposition 4.6(b), which exceeds 1 by a bounded factor. We have not found a proof of (4.3) for all N; an equivalent probabilistic form is that the probability that the d extra cone edges are needed to connect $C_N(h)$ is at most $(q + q^2 + \cdots + q^d)\, f_N(h - d)$.

Proof of Theorem 1.1. Part (i) is Proposition 2.1 of Brown and McMullin (2023). Part (ii) is Theorem 3.3. Part (iii) is Theorem 4.2 and part (iv) is Theorem 4.3; its proviso holds for $7 \le n \le 51$ by Corollary 4.7, which gives the final sentence for these n. □

4.4 Analytic intervals valid at every order

The finite coefficient certificates and the following analytic comparison provide different kinds of coverage. The certificates cover the whole probability interval at bounded orders. The analytic comparison applies at every order, but leaves an intermediate probability interval unresolved.

Theorem 4.8 (Cone comparison on explicit probability intervals)**.** *Let $N \ge 7$, $3 \le s < h \le N - 1$, and $d = h - s$. Then*

$$(1 - q^{d+1}) f_N(s; p) \ge p f_N(h; p), \qquad q = 1 - p,$$

for every $p \in [1/2, 1]$, with strict inequality for $1/2 \le p < 1$.

In the particular range $h = N - 2 - \varepsilon$, $\varepsilon \in \{0, 1\}$ and $1 \le d \le N - 5 - \varepsilon$, put

$$R = \frac{(d+1)(h-d)}{h} \left(\frac{N-1}{N}\right)^d, \qquad L = \binom{N-1}{2} - 1 - \varepsilon.$$

Then $R > 1$, and the same comparison holds on

$$0 \le p \le 1 - R^{-1/L}.$$

Proof. Write $M = N - 1$, $r_M(p) = \mathrm{Rel}(K_M; p)$ and $A = 1 + q + \cdots + q^d$. A sufficient event for the smaller cone to be connected is that the surviving clique is connected and at least one of its s cone edges survives. These events use disjoint edge sets. Conversely, connectivity of the larger cone requires at least one of its h cone edges to survive. Hence

$$f_N(s; p) \ge (1 - q^s) r_M(p), \qquad f_N(h; p) \le 1 - q^h.$$

For $0 < q < 1$ it therefore suffices to prove

$$1 - r_M(p) \le D := 1 - \frac{1 - q^{s+d}}{A(1 - q^s)} = q \frac{1 - q^d}{1 - q^{d+1}} \frac{1 - q^{s-1}}{1 - q^s}.$$

For fixed q, the sequence $(1-q^a)/(1-q^{a+1})$ increases with the positive integer a: the difference of adjacent cross products is $q^a(1-q)^2 > 0$. Since $d \geq 1$ and $s-1 \geq 2$,

$$D \geq q\frac{1-q}{1-q^2}\frac{1-q^2}{1-q^3} = \frac{q}{1+q+q^2}.$$

Every disconnected clique state has a nontrivial cut all of whose edges have failed. Counting each unordered cut once, for $M = 6$ the union bound gives

$$1 - r_6(p) \leq 6q^5 + 15q^8 + 10q^9.$$

If $0 < q \leq 1/2$, division by q bounds this by

$$\frac{6}{16} + \frac{15}{128} + \frac{10}{256} = \frac{17}{32} < \frac{4}{7} \leq \frac{1}{1+q+q^2}.$$

For $M \geq 7$, singleton cuts contribute Mq^{M-1}; each remaining cut has at least $2(M-2)$ crossing edges, and there are fewer than 2^{M-1} such cuts. Thus

$$\frac{1-r_M(p)}{q} \leq Mq^{M-2} + 2^{M-1}q^{2M-5} \leq \frac{M}{2^{M-2}} + 2^{4-M} \leq \frac{7}{32} + \frac{1}{8} = \frac{11}{32} < \frac{4}{7}.$$

Both terms in the penultimate expression decrease with $M \geq 7$. This proves the strict sufficient inequality for $0 < q \leq 1/2$. At $p = 1$ both sides of the cone comparison equal 1.

For the lower interval, the smaller cone has $E_s = \binom{M}{2} + s$ edges. States consisting of exactly a spanning tree give the lower bound

$$f_N(s;p) \geq \tau(C_N(s))p^{N-1}q^{E_s-N+1}.$$

The union bound over spanning trees gives

$$f_N(h;p) \leq \tau(C_N(h))p^{N-1}.$$

Together with $1 - q^{d+1} \geq (d+1)pq^d$, these imply the desired comparison whenever $Rq^L \geq 1$, since

$$R = \frac{(d+1)\tau(C_N(s))}{\tau(C_N(h))}, \qquad d + E_s - N + 1 = \binom{M}{2} - 1 - \varepsilon = L.$$

The cone tree formula yields the stated expression for R.

To verify that the lower interval has positive length, use $s \geq 3$ and $N - 1 \geq d + 4$ to obtain

$$\log R \geq \log\frac{3(d+1)}{d+3} - \frac{d}{d+4} =: \varphi(d),$$

where $\log(1-1/N) \geq -1/(N-1)$ was used. The derivative

$$\varphi'(d) = \frac{2}{(d+1)(d+3)} - \frac{4}{(d+4)^2}$$

has the sign of $10 - d^2$, so φ has only one maximum on $[1,\infty)$. Its value at 1 is $\log(3/2) - 1/5 > 0$ and its limit at infinity is $\log 3 - 1 > 0$. Therefore $R > 1$. The condition $Rq^L \geq 1$ is exactly $p \leq 1 - R^{-1/L}$; at $p = 0$ the comparison holds directly. □

In particular, the threshold-graph reduction applies pointwise: for $\varepsilon \in \{0,1\}$, $n \geq 8+3\varepsilon$ and $m = \binom{n-1}{2} - \varepsilon$, every connected graph H with n vertices and m edges satisfies

$$\mathrm{Rel}(H;p) \geq \mathrm{Rel}(B_{n,m};p) \qquad (1/2 \leq p \leq 1).$$

The same implication holds on the common lower interval obtained by taking the minimum of $1-R^{-1/L}$ over the finitely many required values of d for $N = n-1$. These are pointwise interval conclusions, not uniform reliability on $[0,1]$. The intermediate interval remains unresolved at unrestricted orders; the full-interval finite theorem through $n = 51$ remains a separate result.

5 Exact computations and their scope

The universal nonexistence and dense-existence arguments above are written proofs. The finite boundary conclusion additionally depends on exact cone coefficients and a small number of all-terminal base cases. The other enumerations are corroboration, not substitutes for those arguments. None of these programs is a proof-assistant formalisation of the mathematics.

5.1 Certificate semantics

For each graph, c_i counts connected spanning edge sets of size i, and N_i counts the corresponding split states. The connected-subgraph generating polynomial on an induced vertex set S is obtained by subtracting states whose component containing the least vertex is a proper subset T:

$$c_S(x) = (1+x)^{e(S)} - \sum_{\substack{T \subsetneq S \\ \min S \in T}} c_T(x)(1+x)^{e(S \setminus T)}.$$

Edges between T and its complement are absent. Split coefficients then come from

$$\sum_{\substack{A \subseteq V \\ s \in A,\ t \notin A}} c_A(x) c_{V \setminus A}(x).$$

Counting every terminal pair does not require treating isomorphic copies as different scientific objects. The graph6 corpora contain unlabelled graph representatives; all their unordered terminal pairs are checked when a full split-vector run is specified.

A coefficientwise inequality at a common total edge count implies a probability inequality, because $p^i(1-p)^{m-i} \geq 0$. It is sufficient, not necessary. Nonexistence is certified differently: the graphs maximising the lowest split coefficient and those maximising the reversed coefficient vector lexicographically have disjoint sets of coefficient vectors. The endpoint lemma then rules out a uniform optimiser. Failure of a coefficientwise comparison alone is not such a nonexistence certificate.

5.2 Cone coefficients

Write $M = N-1$. If a of the s cone edges survive, the surviving clique must have every component meeting their a endpoints. Let $\gamma_M(a;x)$ count those clique edge sets. Removing the union of components avoiding that fixed root set gives

$$\gamma_M(a;x) = (1+x)^{\binom{M}{2}} - \sum_{u=1}^{M-a} \binom{M-a}{u} (1+x)^{\binom{u}{2}} \gamma_{M-u}(a;x),$$

and hence

$$\sum_j c_j(C_N(s))x^j = \sum_{a=1}^{s} \binom{s}{a} x^a \gamma_M(a;x).$$

The fast implementation uses packed integer polynomials with an explicit digit bound; an alternative implementation groups the generic subset recursion by symmetry orbits. Ordinary-list arithmetic and direct edge-subset checks provide smaller, transparent comparisons. Packing arithmetic is an implementation choice, not a strengthening of the claim.

5.3 Recorded coverage

The incoming reviewer evidence records the following executed checks. These are supplied review provenance, not executions newly attributed to this revision. The release's replay receipt separately identifies executions performed on the revised package and their exact input hashes.

Check	Exact scope	Role
Required cone inequalities	$7 \le N \le 50$, both ε, every required d: 2,024 triples and 1,354,694 coefficient comparisons	Finite boundary theorem
Boundary all-terminal minima	Both boundary edge counts at $n = 7, 8, 9, 10$; 5,422 graphs	Base cases for the reduction
Rebuilt C split vectors	All connected graphs at $n = 5, 6, 7, 8$, and the supplied dense $n = 10$ corpus; 16,678 graphs, 536,954 terminal pairs	Implementation replay
Direct edge-subset comparison	All 133 connected graphs at $n = 5, 6$; 1,890 terminal pairs	Independent algorithm check within supplied review
Root-set and generic cone checks	21 root-set checks through clique order 6; 39 cone checks at orders 5–10	Formula and implementation check
Additional dense $n = 9$ replay	1,370 graphs, 49,320 terminal pairs	Supplementary corroboration
Corpus regeneration	Nine supplied graph6 corpora	Input enumeration check
First excluded cone value	88 cases: 83 leading-coefficient failures and five full coefficientwise passes	Correction of false sharpness assertion

The original identity/identified-graph logs cover **3,640 terminal pairs from 140 graph representatives**: 40 dense graphs of order 7 and 100 of order 8. They do not cover every two-terminal graph through order 8. For adjacent terminals the loop-padded coefficient identity from the preliminaries is required.

The original fast cone logs reach $N = 50$. Complete two-ε coverage of the slower implementation ends at $N = 30$, with only $\varepsilon = 0$ recorded at $N = 31$. Its generic validation covers $N = 5, 6, 7$; the faster implementation's validation covers $N = 5, \ldots, 10$. We do not describe these as two complete independent runs through $N = 50$.

Original summaries also report full split enumeration at $n = 9$ and dense $n = 11, 12$. Their large raw split files were not included in the source handoff, and the supplied review did not replay all that arithmetic. They are retained as historical supplementary records, not additional dependencies of the finite boundary theorem. Winner summaries are not certificates of all terminal-pair isomorphism classes. Likewise, a pendant-plus-core description applies only where the edge count permits a pendant vertex, not to the entire dense range.

5.4 A small certificate readers can inspect

At $N = 7$, $\varepsilon = 1$, the first excluded value is $d = 2$. This is one of the five exceptions to the original sharpness assertion. With $q = 1 - p$,

$$\Delta(p) = (1 - q^3)\,\mathrm{Rel}(C_7(2); p) - p\,\mathrm{Rel}(C_7(4); p) = \sum_{j=0}^{20} D_j p^j q^{20-j}.$$

Only $D_7, \ldots, D_{19}$ are nonzero, and they are

$$840,\ 7640,\ 19904,\ 30348,\ 32615,\ 26418,\ 16510,$$

$$7976,\ 2938,\ 800,\ 152,\ 18,\ 1.$$

They are all positive, so $\Delta(p) > 0$ throughout $0 < p < 1$; at $p = 1/2$ it is $9135/65536$. A standalone enumerator checks all $2^{17} + 2^{19} = 655360$ edge sets without importing the submitted graph routines. This example also explains why excluded parameter values must be tested explicitly rather than inferred from a loop over the required positive range.

5.5 Reproducibility boundary

The package's claim-to-file map gives the exact scripts, corpus files and logs. Required checks return nonzero on a failed comparison, and negative controls exercise the loop convention and incorrect boundary assertions. Historical logs are not rewritten. Rebuild the C implementation from source; bundled platform-specific executables are not part of the public evidence contract. The integer bounds apply only in the documented supported range. Reproducing outputs checks the implemented finite statement, not the imported literature or every step of the universal prose proof.

6 Discussion

The threshold $m = \binom{n}{2} - n$ has a transparent meaning in the light of Lemma 2.3: a split event is a connectivity event on the identified graph H/st minus a connectivity event on H itself, and the balloon candidate is simultaneously the graph whose identification is most reliable (Lemma 4.1, true for all $k \le n$) and the graph that is itself least reliable (Petingi–Saccoman–Schoppmann, known for $k \le n - 2$). For $k \ge n + 1$ the first property fails to compensate: the lexicographically forced candidate is beaten at the 2-forest coefficient by graphs whose missing edges are spread rather than concentrated, as Section 3 shows with explicit spanning-tree counts.

The two edge counts left conditional, $m = \binom{n-1}{2}$ and $m = \binom{n-1}{2} - 1$, are exactly the first two below the range in which the all-terminal "uniformly least reliable" problem is solved. The statement that would complete the characterisation is the following instance of the conjecture of Boesch, Satyanarayana and Suffel.

Conjecture 6.1. *Let $n \geq 7$ and let H be a connected graph with n vertices and m edges.*

(a) If $m = \binom{n-1}{2}$, then $\mathrm{Rel}(H;p) \geq p\,\mathrm{Rel}(K_{n-1} - e;p)$ for all $p \in [0,1]$, where $K_{n-1} - e$ is the complete graph with one edge removed.

(b) If $m = \binom{n-1}{2} - 1$, then $\mathrm{Rel}(H;p) \geq p\,\mathrm{Rel}(K_{n-1} - P_3;p)$ for all $p \in [0,1]$, where $K_{n-1} - P_3$ is the complete graph with two adjacent edges removed.

The right-hand sides are the reliabilities of the balloon graphs $B_{n,m}$ at the two edge counts (a pendant vertex attached anywhere to the core contributes the factor p, so the choice of attachment vertex is immaterial). By Theorem 4.3, Conjecture 6.1 implies that a UMSR graph exists in $\mathcal{T}_{n,m}$ if and only if $m = n-1$ or $m \geq \binom{n}{2} - n$, for every $n \geq 7$. Theorem 4.5 shows that Conjecture 6.1 for $n \geq 11$ follows from the inequality $(\star)_{n-1,\varepsilon}$ between the two cone graphs $C_{n-1}(n-3-\varepsilon)$ and $C_{n-1}(n-3-\varepsilon-d)$, and Corollary 4.7 establishes it for $7 \leq n \leq 51$. A sufficient remaining research target is therefore a family indexed by N, ε, d, uniformly quantified over p:

Conjecture 6.2. *$(\star)_{N,\varepsilon}$ holds for every $N \geq 7$ and $\varepsilon \in \{0,1\}$; that is, for $h = N-2-\varepsilon$ and $1 \leq d \leq N-5-\varepsilon$,*

$$(1-q^{d+1})\,\mathrm{Rel}(K_N - K_{1,1+\varepsilon+d};p) \;\geq\; p\,\mathrm{Rel}(K_N - K_{1,1+\varepsilon};p) \qquad (0 \leq p \leq 1).$$

Conjecture 6.2 implies Conjecture 6.1 and hence completes the characterisation for all $n \geq 7$. It holds coefficient-wise for $N \leq 50$ (Proposition 4.6). Theorem 4.8 proves it at every order for $p \geq 1/2$ and on an explicit interval near zero; only the intervening interval remains to be proved at unrestricted orders. The first excluded parameter value has a leading-coefficient obstruction for $N \geq 11$, so a proof cannot simply extend the required range without qualification.

The theorem classifies existence, not all terminal-preserving isomorphism classes of optimisers. Romero's theorem identifies the optimisers, whenever they exist, by split-equivalence to the balloon polynomial. The supplied winner summaries retain only representative terminal pairs and do not certify an enumeration of every terminal-placement class. We make no new terminal-isomorphism uniqueness claim on that evidence.

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